

Hölder Reflected-Resolvent Geometry at Multivalued Branch Junctions: Error Bounds and Proximal Point Convergence

菠萝

Abstract

Local convergence analyses of the proximal point algorithm commonly rely on monotonicity or on quadratic generalized-monotonicity conditions that make the reflected resolvent locally Lipschitz. Such a description can break down at a junction of branches of a set-valued operator: the resolvent may remain single-valued on the relevant Minty image while its reflection is only Hölder continuous across the junction. We study this regime through the all-pairs graph condition

$$\|(u - v) - \lambda(u^* - v^*)\| \leq L\|(u - v) + \lambda(u^* - v^*)\|^\gamma, \quad 0 < \gamma \leq 1,$$

imposed on a local graph block of an operator F . In Minty coordinates, this condition is exactly a Hölder estimate for the reflected resolvent; its all-pairs quantifier retains comparisons both within and between active branches. We combine this geometry with an independently verified error bound of the form

$$\text{dist}(y, S) \leq \psi(\text{dist}(0, F(y))), \quad S = F^{-1}(0),$$

and with local resolvent coverage. Two explicit scalar compatibility branches then yield a strict Lyapunov contraction; in the power-law cases this becomes a quantitative contraction of the distance to S . A separate step estimate gives finite length, keeps the iterates inside the verified region, and proves convergence of the whole orbit to a point of the possibly nonisolated solution set. We also give a verification route based on native branch data: within-branch estimates, cross-branch junction estimates, absence of Minty collisions, coverage, and an output error bound are checked separately before being assembled. A square-root junction example has optimal exponent $\gamma = 1/2$ and admits no local Lipschitz reflected-resolvent bound, while the nonlinear geometry and its matching error bound certify the proximal iteration. Thus the framework extends the usable operator geometry beyond fixed quadratic certificates without replacing the independent regularity and coverage assumptions needed for convergence.

1 Introduction

Let $F : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ and $\lambda > 0$. The proximal point iteration selects

$$x^{k+1} \in J_{\lambda F}(x^k), \quad J_{\lambda F} := (I + \lambda F)^{-1}, \quad (1)$$

with the aim of reaching the zero set $S := F^{-1}(0)$. For monotone operators, firm nonexpansiveness of the resolvent supplies both a stable one-step geometry and the basic descent mechanism behind the classical theory [15]; for nonmonotone operators, related analyses replace monotonicity by submonotonicity, hypomonotonicity, semimonotonicity, or almost-averaged properties, and combine that geometry with metric subregularity or another error bound [11, 12, 2].

The quadratic nature of these conditions is well suited to locally Lipschitz resolvents. It is less well suited to a genuinely multivalued graph near a branch junction. To see the obstruction, write the Minty input and reflected output associated with a graph point $(u, u^*) \in \text{gph } F$ as

$$x = u + \lambda u^*, \quad q = u - \lambda u^*. \quad (2)$$

At a junction, two nearby inputs can lie on different branches. Even when the complete inverse relation selects one output for every relevant input, the cross-branch estimate may have the scale

$$\|q - q'\| \asymp \|x - x'\|^{1/2}. \quad (3)$$

Consequently,

$$\frac{\|q - q'\|}{\|x - x'\|} \asymp \|x - x'\|^{-1/2} \longrightarrow +\infty,$$

so no finite local Lipschitz constant—and hence no fixed quadratic reflected-resolvent certificate of this type—can describe the complete junction. Checking each smooth branch separately does not repair the problem: the missing comparison is precisely the one between different branches.

This paper records the junction scale directly. For two graph points $(u, u^*), (v, v^*) \in \text{gph } F$, set

$$a = u - v, \quad b = u^* - v^*.$$

We impose the local all-pairs condition

$$\|a - \lambda b\| \leq L\|a + \lambda b\|^\gamma, \quad L \geq 0, \quad 0 < \gamma \leq 1. \quad (4)$$

We call (4) the *RL condition*. Under the Minty transformation (2), it is exactly the Hölder estimate

$$\|Q_{\lambda F}x - Q_{\lambda F}y\| \leq L\|x - y\|^\gamma, \quad Q_{\lambda F} := 2J_{\lambda F} - I,$$

on the relevant Minty image. The formulation on $\text{gph } F$ is important: it makes the all-pairs requirement explicit, detects a Minty collision by setting $a + \lambda b = 0$, and does not silently discard a branch by choosing a convenient local selection. The case $\gamma = 1$ is a quadratic slice connected with established generalized-monotonicity notions; the regime $0 < \gamma < 1$ permits a nonquadratic junction geometry. Merely writing a Lipschitz map with a smaller Hölder exponent is not the phenomenon of interest. In the examples below, the exponent is optimal and local Lipschitz continuity fails.

The RL condition alone does not force a proximal orbit toward S . It controls how the complete graph is seen in Minty coordinates, not how an output residual separates a point from the zero set. We therefore keep a second hypothesis independent:

$$\text{dist}(y, S) \leq \psi(\text{dist}(0, F(y))), \quad (5)$$

where ψ is a nondecreasing gauge with $\psi(0) = 0$ and $\psi(t) \rightarrow 0$ as $t \downarrow 0$. Conditions (4) and (5) answer different questions: RL measures the tangential and cross-branch distortion of the Minty graph, whereas the error bound measures the normal recovery of the solution set from the residual. Their constants and orders are verified separately and are coupled only at the final scalar step of the convergence proof.

Indeed, let $x^+ \in J_{\lambda F}(x)$, choose $p \in P_S(x)$, and put

$$d = \text{dist}(x, S), \quad d^+ = \text{dist}(x^+, S), \quad s = \|x - x^+\|.$$

Comparison of $(x^+, (x - x^+)/\lambda)$ with $(p, 0)$ in (4) gives the two basic estimates

$$(d^+)^2 + s^2 \leq \frac{1}{2}(d^2 + L^2 d^{2\gamma}), \quad (6)$$

$$s \leq \frac{1}{2}(d + Ld^\gamma). \quad (7)$$

On the other hand, the proximal inclusion and (5) yield

$$d^+ \leq \psi(s/\lambda). \quad (8)$$

The coupled inequalities (6) and (8), or the sharper branch selected in the main theorem, provide a strict scalar Lyapunov contraction under an explicit compatibility condition. Estimate (7) then has a different role: the geometrically decaying Lyapunov values control a summable step sequence. This proves finite length, supplies the localization budget needed to reuse the local assumptions, and upgrades convergence to the set into convergence of the entire orbit to one point of S . Keeping these two roles separate prevents a one-step Lyapunov estimate from being mistaken for a complete local PPA theorem.

The practical issue is verification. For a multivalued operator one rarely knows an exact global formula for its graph. More often one knows active branches, local constitutive laws, growth orders, generalized derivatives, or an independently established error bound. Our elementary assembly scheme therefore uses a complete local branch atlas. It checks within-branch estimates, cross-branch junction estimates, and the absence of Minty collisions on the full relevant graph block. Coverage of the Minty input neighborhood is kept as a separate hypothesis; injectivity of the graph transform does not imply coverage. Finally, an error-bound certificate is supplied for the same output points and the same solution set used by the proximal transition. This modular organization allows existing generalized-differentiation and error-bound tools to be imported without asking one theorem to produce every part of the certificate [14, 1, 6, 13, 9].

Contributions

The paper makes the following bounded contributions.

1. **Two explicit convergence certificates and a whole-orbit theorem.** We combine RL geometry, the independent gauge error bound (5), and local resolvent coverage through an explicit scalar compatibility condition. The step-bound and energy branches are generally incomparable; at the nonlinear critical order they give the distinct scalar thresholds $L/2$ and $L/\sqrt{2}$, respectively. The conclusion includes strict Lyapunov decay, finite length, localization of every iterate, and convergence to an individual point of a possibly nonisolated solution set. For power error bounds this yields explicit distance factors and exposes the precise comparison between the RL order and the residual-growth order.
2. **A graph-level nonlinear reflected-resolvent condition.** We formulate (4) on a complete local graph block and establish its equivalence with an all-pairs Hölder estimate for the reflected resolvent on the associated Minty image. The formulation distinguishes single-valuedness on the image from coverage of a neighborhood and retains cross-branch comparisons.
3. **Elementary assembly rules from native branch information.** We separate the certificate into within-branch regularity, cross-branch seam control, Minty injectivity,

local coverage, and output error bounds. The resulting rules convert these independently checkable ingredients into usable RL and regularity parameters while making explicit which part of the argument fails if a branch or an active seam is omitted.

4. **A strict nonlinear junction example.** We construct a multivalued example with a nonisolated solution set whose complete reflected resolvent has optimal exponent $1/2$. A sequence comparing the two sides of the seam proves the failure of every finite local Lipschitz bound and hence of the corresponding fixed quadratic certificate, while a matching error bound and explicit constants place the example inside the convergence theorem. The example separates nonlinear junction geometry from an artificial weakening of a Lipschitz estimate.
5. **A precise relation to the linear slice.** For $\gamma = 1$, we identify the RL inequality with the appropriate scalar quadratic semimonotonicity slice and record its relation to monotonicity, hypomonotonicity, and submonotonicity-based PPA analyses. We claim an extension in operator geometry only where the reflected resolvent is genuinely non-Lipschitz; no claim of dominance over general Lyapunov, partial-regularity, or model-specific convergence arguments is made.

Relation to existing theory. The linear case belongs to the established family of quadratic generalized monotonicity conditions [3, 2, 11]. Metric and Hölder subregularity, including higher-order and gauge variants, also have independent mature theories [6, 13, 9]. Our contribution is not either ingredient in isolation. It is the quantitative pairing of a full-graph non-Lipschitz Minty geometry with an independently verified output error bound, together with a branch-complete verification route and a whole-orbit localization argument. Conversely, an error bound cannot repair a Minty collision, and RL injectivity cannot supply a missing input. These boundaries are retained throughout the paper.

Organization. Section 2 develops the RL geometry, the two scalar compatibility branches, and the local whole-orbit theorem. Section 3 gives the native branch-atlas assembly rules and analyzes the square-root junction, including the lower-bound sequence proving exponent optimality. Section 4 states the exact relations and nonrelations with the closest generalized-monotonicity and regularity frameworks. Section 5 summarizes the contribution and the remaining boundary.

2 Reflected-resolvent geometry and error bounds

Throughout this section, $X = \mathbb{R}^n$ is equipped with its Euclidean norm, $F : X \rightrightarrows X$, and $\lambda > 0$. For a graph point $(u, u^*) \in \text{gph } F$, introduce the Minty coordinates

$$M_\lambda^+(u, u^*) := u + \lambda u^*, \quad M_\lambda^-(u, u^*) := u - \lambda u^*.$$

All statements below are local and may be imposed on a specified graph block $\mathcal{G} \subset \text{gph } F$.

Definition 2.1 (Graph-level RL condition). Let $\omega : [0, R] \rightarrow [0, \infty)$ be nondecreasing, with $\omega(0) = 0$. The graph block $\mathcal{G}$ satisfies $\text{RL}(\lambda, \omega)$ at scale R if, for every $(u, u^*), (v, v^*) \in \mathcal{G}$ such that $\|M_\lambda^+(u, u^*) - M_\lambda^+(v, v^*)\| \leq R$,

$$\|M_\lambda^-(u, u^*) - M_\lambda^-(v, v^*)\| \leq \omega(\|M_\lambda^+(u, u^*) - M_\lambda^+(v, v^*)\|). \quad (9)$$

The power-law specialization is $\omega(t) = Lt^\gamma$, where $L \geq 0$ and $0 < \gamma \leq 1$.

Lemma 2.2 (Minty representation and the coverage distinction). *Suppose that $\mathcal{G}$ satisfies (9). Then M_λ^+ is injective on $\mathcal{G}$. Consequently the maps*

$$J_{\mathcal{G}}(x) := u, \quad Q_{\mathcal{G}}(x) := u - \lambda u^*, \quad x = u + \lambda u^*,$$

are single-valued on the Minty image $\mathcal{M}_{\mathcal{G}} := M_\lambda^+(\mathcal{G})$, and

$$\|Q_{\mathcal{G}}x - Q_{\mathcal{G}}y\| \leq \omega(\|x - y\|), \quad x, y \in \mathcal{M}_{\mathcal{G}}, \quad \|x - y\| \leq R. \quad (10)$$

Conversely, (10) on $\mathcal{M}_{\mathcal{G}}$ is equivalent to (9). Moreover,

$$\|J_{\mathcal{G}}x - J_{\mathcal{G}}y\|^2 + \|(I - J_{\mathcal{G}})x - (I - J_{\mathcal{G}})y\|^2 \leq \frac{\|x - y\|^2 + \omega(\|x - y\|)^2}{2}. \quad (11)$$

None of these conclusions implies that $\mathcal{M}_{\mathcal{G}}$ contains a neighborhood of a prescribed input. Thus Minty injectivity, pointwise resolvent existence, and neighborhood coverage are logically separate.

Proof. If two graph points have the same M_λ^+ coordinate, then (9) and $\omega(0) = 0$ give equality of their M_λ^- coordinates. Adding and subtracting the two coordinate identities gives equality of both graph points. This proves injectivity. The identity (10) is merely (9) written in Minty coordinates, so the converse is immediate. Finally,

$$J_{\mathcal{G}} = \frac{I + Q_{\mathcal{G}}}{2}, \quad I - J_{\mathcal{G}} = \frac{I - Q_{\mathcal{G}}}{2},$$

and the parallelogram identity gives

$$\|J_{\mathcal{G}}x - J_{\mathcal{G}}y\|^2 + \|(I - J_{\mathcal{G}})x - (I - J_{\mathcal{G}})y\|^2 = \frac{\|x - y\|^2 + \|Q_{\mathcal{G}}x - Q_{\mathcal{G}}y\|^2}{2}.$$

The final assertion is witnessed, for example, by the graph consisting only of $(0, 0)$: it satisfies every such inequality, while its Minty image is the singleton $\{0\}$. $\square$

Let $S \subset F^{-1}(0)$ be nonempty and closed. Given an input x and a proximal point $x^+ \in J_{\lambda F}(x)$, set

$$d := d(x, S), \quad d^+ := d(x^+, S), \quad s := \|x - x^+\|.$$

For later use define

$$A(t) := \frac{t^2 + \omega(t)^2}{2}, \quad B(t) := \frac{(t + \omega(t))^2}{4}. \quad (12)$$

Notice that $B(t) \leq A(t)$, but the two criteria below divide these quantities by different Lyapunov gauges; neither resulting criterion dominates the other in general.

Lemma 2.3 (Solution comparison). *Assume that the graph point $(x^+, (x - x^+)/\lambda)$ and every $(p, 0)$ with $p \in P_S(x)$ belong to $\mathcal{G}$, and that (9) is applicable at distance d . Then*

$$(d^+)^2 + s^2 \leq A(d), \quad (13)$$

$$s^2 \leq B(d). \quad (14)$$

Proof. Fix $p \in P_S(x)$ and put $a = x^+ - p$ and $b = (x - x^+)/\lambda$. Then $a + \lambda b = x - p$, whereas (9) gives $\|a - \lambda b\| \leq \omega(d)$. The parallelogram identity yields

$$\|x^+ - p\|^2 + s^2 = \frac{\|a + \lambda b\|^2 + \|a - \lambda b\|^2}{2} \leq A(d).$$

Since $d^+ \leq \|x^+ - p\|$, this proves (13). Also,

$$2s = \|(a + \lambda b) - (a - \lambda b)\| \leq d + \omega(d),$$

which proves (14). $\square$

Definition 2.4 (Admissible error-bound gauge). An admissible gauge on $[0, \bar{t}]$ is a continuous, strictly increasing map $\psi : [0, \bar{t}] \rightarrow [0, \infty)$ with $\psi(0) = 0$. Its inverse is understood on $[0, \psi(\bar{t})]$. We say that F obeys the ψ -error bound relative to S on a set V if

$$d(y, S) \leq \psi(d(0, F(y))), \quad y \in V, \quad d(0, F(y)) \leq \bar{t}. \quad (15)$$

Define the induced Lyapunov gauge

$$h(r) := \lambda^2 [\psi^{-1}(r)]^2, \quad 0 \leq r \leq \psi(\bar{t}). \quad (16)$$

The same arguments remain valid for a merely nondecreasing gauge when $\psi^{-1}(r) := \inf\{t : r \leq \psi(t)\}$ is positive for $r > 0$ and the resulting h separates zero; the class above avoids irrelevant inverse pathologies.

Lemma 2.5 (Error bound along a proximal transition). *Suppose that (15) holds at x^+ and that $s/\lambda \leq \bar{t}$. Then*

$$h(d^+) \leq s^2. \quad (17)$$

Proof. The proximal inclusion gives $(x - x^+)/\lambda \in F(x^+)$ and therefore $d(0, F(x^+)) \leq s/\lambda$. Monotonicity of ψ in (15) gives $d^+ \leq \psi(s/\lambda)$. Applying ψ^{-1} and using (16) proves (17). $\square$

We next state all local assumptions explicitly. Let $U \subset X$ be open and let $R > 0$. Assume:

- (L1) $S \subset F^{-1}(0)$ is nonempty and closed, and $S \times \{0\} \subset \mathcal{G}$.
- (L2) The block $\mathcal{G}$ satisfies $\text{RL}(\lambda, \omega)$ for Minty-input separations at most R .
- (L3) For every $x \in U$ with $d(x, S) \leq R$, the full Minty fiber $\{(u, u^*) \in \text{gph } F : u + \lambda u^* = x\}$ is contained in $\mathcal{G}$ and is nonempty. Thus this is both a coverage and an input-completeness assumption; by Lemma 2.2 the ordinary resolvent $J_{\lambda F}(x)$ is a singleton there.
- (L4) The error bound (15) holds at every output $J_{\lambda F}(x)$ generated from the preceding inputs.
- (L5) $R \leq \psi(\bar{t})$ and $(R + \omega(R))/(2\lambda) \leq \bar{t}$.

Assumption (L3) is not a consequence of (L2).

Theorem 2.6 (Two deterministic RLEB convergence branches). *Suppose assumptions (L1)–(L5) hold. Assume that at least one of the following conditions holds for some $q \in (0, 1)$:*

$$q_H(R) := \sup_{0 < r \leq R} \frac{B(r)}{h(r)} \leq q < 1, \quad (\text{H})$$

$$q_E(R) := \sup_{0 < r \leq R} \frac{A(r)}{r^2 + h(r)} \leq q < 1. \quad (\text{E})$$

Let $x^0 \in U$, put $d_0 = d(x^0, S) \leq R$, and define

$$\mathcal{L}_H(d_0) := \frac{\sqrt{q h(d_0)}}{1 - \sqrt{q}}, \quad \mathcal{L}_E(d_0) := \frac{\sqrt{q [d_0^2 + h(d_0)]}}{1 - \sqrt{q}}. \quad (18)$$

Use $\mathcal{L}_H$ under (H) and $\mathcal{L}_E$ under (E). If the corresponding localization condition

$$\mathcal{L}_\bullet(d_0) < \text{dist}(x^0, X \setminus U) \quad (19)$$

holds, then the proximal iteration

$$x^{k+1} = J_{\lambda F}(x^k) \quad (20)$$

is well defined for every k , remains in $U \cap \{x : d(x, S) \leq R\}$, has finite length, and converges to a point $x^\infty \in S$.

More precisely, with $d_k = d(x^k, S)$ and $s_k = \|x^{k+1} - x^k\|$, branch (H) gives

$$h(d_k) \leq q^k h(d_0), \quad (21)$$

$$s_k \leq q^{(k+1)/2} \sqrt{h(d_0)}, \quad (22)$$

$$\|x^\infty - x^k\| \leq \frac{q^{(k+1)/2} \sqrt{h(d_0)}}{1 - \sqrt{q}}. \quad (23)$$

Branch (E), with $V(r) := r^2 + h(r)$, gives

$$V(d_k) \leq q^k V(d_0), \quad (24)$$

$$s_k \leq q^{(k+1)/2} \sqrt{V(d_0)}, \quad (25)$$

$$\|x^\infty - x^k\| \leq \frac{q^{(k+1)/2} \sqrt{V(d_0)}}{1 - \sqrt{q}}. \quad (26)$$

Thus the theorem supplies quantitative pointwise tails, not only decay of the distance to the solution set. Here and below $\text{dist}(x^0, X \setminus U) := +\infty$ when $U = X$.

Proof. Assume first that an iterate x^k lies in $U \cap \{x : d(x, S) \leq R\}$. Assumption (L3) and Lemma 2.2 give a unique proximal point x^{k+1} . Assumptions (L1)–(L2) allow comparison with a projection of x^k onto S , so Lemma 2.3 applies. Since ω is nondecreasing, (14) and (L5) place the selected residual inside the error-bound range. Lemma 2.5 therefore gives

$$d_{k+1}^2 + s_k^2 \leq A(d_k), \quad s_k^2 \leq B(d_k), \quad h(d_{k+1}) \leq s_k^2. \quad (27)$$

Under branch (H), (27) and (H) yield

$$h(d_{k+1}) \leq s_k^2 \leq B(d_k) \leq q h(d_k).$$

Iteration gives (21), and substituting it back gives (22). Since h is strictly increasing, $d_{k+1} \leq d_k \leq R$.

Under branch (E), (27) gives

$$V(d_{k+1}) = d_{k+1}^2 + h(d_{k+1}) \leq d_{k+1}^2 + s_k^2 \leq A(d_k) \leq qV(d_k).$$

This proves (24); moreover $s_k^2 \leq A(d_k) \leq qV(d_k)$ gives (25). Because V is strictly increasing, again $d_{k+1} \leq d_k \leq R$.

In either branch the partial path length through step N is no larger than the corresponding geometric sum, hence no larger than $\mathcal{L}_\bullet(d_0)$. The strict localization condition (19) therefore keeps x^N inside U . Induction gives existence of the entire orbit and validates every use of the local assumptions. Summing from k to infinity proves (23) or (26), as well as finite length. Hence (x^k) is Cauchy and converges to some x^∞ . The Lyapunov decay implies $d_k \rightarrow 0$, and closedness of S gives $x^\infty \in S$. $\square$

Remark 2.7 (What is and is not sharp). Each ratio in (H)–(E) is the exact uniform constant delivered by its displayed scalar implication. The two branches use different information: (H) combines the step estimate with the error bound, while (E) retains the coupled energy inequality. Neither condition is asserted to be necessary for PPA convergence, and neither constant is asserted to be the optimal contraction factor over all operators satisfying graph-level RL. In particular, calling the threshold obtained from branch (E) the sharp threshold of the complete RL–PPA hypotheses would be incorrect because branch (H) may certify strict contraction when branch (E) does not.

The two branches are already incomparable in the linear slice. With $\lambda/\rho = 1$ and $L = 6/5$, branch (E) holds while branch (H) fails. With $\lambda/\rho = 3$ and $L = 9/2$, branch (H) holds while branch (E) fails.

2.1 Power-law and linear slices

Corollary 2.8 (Power gauges). *In the setting of Theorem 2.6, let*

$$\omega(r) = Lr^\gamma, \quad \psi(t) = Kt^\theta, \quad L, K > 0, \quad 0 < \gamma \leq 1, \quad \theta > 0,$$

and set

$$c := \lambda^2 K^{-2/\theta}, \quad p := \frac{2}{\theta}.$$

Then $h(r) = cr^p$, and the two computable ratios are

$$q_H(R) = \frac{1}{4c} \sup_{0 < r \leq R} \left(r^{1-1/\theta} + Lr^{\gamma-1/\theta} \right)^2, \quad (28)$$

$$q_E(R) = \sup_{0 < r \leq R} \frac{r^2 + L^2 r^{2\gamma}}{2(r^2 + cr^{2/\theta})}. \quad (29)$$

If $q_H(R) < 1$, then every orbit satisfying the localization hypothesis of Theorem 2.6 obeys the additional distance contraction

$$d_{k+1} \leq q_H(R)^{\theta/2} d_k. \quad (30)$$

If $q_E(R) < 1$, the corresponding distance contraction is

$$d_{k+1} \leq q_E(R)^{1/\max\{2, 2/\theta\}} d_k. \quad (31)$$

Proof. The formulas follow by substituting the two power gauges into (16), (12), (H), and (E). Since $h(d) = cd^{2/\theta}$, the branch-(H) inequality $h(d_{k+1}) \leq q_H(R)h(d_k)$ is equivalent to (30). For branch (E), put $V(r) = r^2 + cr^{2/\theta}$ and $m = \max\{2, 2/\theta\}$. For $0 \leq \kappa \leq 1$, one has $V(\kappa r) \geq \kappa^m V(r)$. Taking $\kappa = q_E(R)^{1/m}$ in the branch-(E) Lyapunov inequality gives (31). $\square$

Corollary 2.9 (Residual growth and the nonlinear critical order). *In the setting of Theorem 2.6, let $0 < \gamma < 1$, $L > 0$, and suppose that, on the relevant output neighborhood,*

$$d(0, F(y)) \geq m d(y, S)^a, \quad m > 0, \quad a > 0. \quad (32)$$

Then one may take $\psi(t) = (t/m)^{1/a}$ and $h(r) = \lambda^2 m^2 r^{2a}$. Consequently:

1. *if $a < \gamma$, then both scalar ratios tend to zero as $R \downarrow 0$, so both branches satisfy the scalar compatibility condition locally;*
2. *if $a = \gamma$, branch (H) certifies the theorem for all sufficiently small R exactly when*

$$\lambda m > \frac{L}{2}, \quad (33)$$

whereas branch (E) requires the stronger condition $\lambda m > L/\sqrt{2}$;

3. *if $a > \gamma$, both ratios diverge as $R \downarrow 0$, so these two scalar certificates do not establish the present contraction theorem. This is a failure of the certificates, not a converse divergence statement about PPA.*

In the first two cases, the orbit conclusions still require the coverage, range, and localization hypotheses of Theorem 2.6; the scalar order comparison does not supply them.

Proof. In Corollary 2.8, use $\theta = 1/a$, $K = m^{-1/a}$, and hence $c = \lambda^2 m^2$. If $a < \gamma$, every leading exponent in (28) is positive, and in (29) the denominator is asymptotically dominated by $2\lambda^2 m^2 r^{2a}$; both ratios tend to zero. If $a = \gamma$, then

$$\lim_{R \downarrow 0} q_H(R) = \frac{L^2}{4\lambda^2 m^2}, \quad \lim_{R \downarrow 0} q_E(R) = \frac{L^2}{2\lambda^2 m^2},$$

which gives the two strict thresholds. If $a > \gamma$, the branch-(H) ratio contains the divergent factor $r^{2(\gamma-a)}$. For branch (E), the denominator has order $r^{2\min\{1, a\}}$, and $\min\{1, a\} > \gamma$ because $\gamma < 1$ and $a > \gamma$; hence its ratio also diverges. $\square$

Corollary 2.10 (The linear slice). *In the setting of Theorem 2.6, let $\gamma = 1$, $\omega(r) = Lr$, and assume the linear error bound*

$$d(y, S) \leq \rho d(0, F(y)), \quad \rho > 0. \quad (34)$$

Then the two certified distance factors are

$$\kappa_H = \frac{(1+L)\rho}{2\lambda}, \quad \kappa_E = \sqrt{\frac{1+L^2}{2}} \frac{\rho}{\sqrt{\rho^2 + \lambda^2}}. \quad (35)$$

Branch (H) applies when $(1+L)\rho < 2\lambda$, and branch (E) applies when

$$L^2 < 1 + \frac{2\lambda^2}{\rho^2}. \quad (36)$$

Whenever either inequality holds, Theorem 2.6 applies to every initial point satisfying its localization hypothesis, with the corresponding squared factor $q = \kappa_H^2$ or $q = \kappa_E^2$. These are sufficient factors furnished by the two branches; no necessity or operator-class-wide optimality is claimed.

Proof. Here $h(r) = \lambda^2 r^2 / \rho^2$. Direct substitution gives $q_H = (1 + L)^2 \rho^2 / (4\lambda^2)$ and

$$q_E = \frac{1 + L^2}{2} \frac{\rho^2}{\rho^2 + \lambda^2}.$$

Taking square roots and solving $q_H < 1$ and $q_E < 1$ proves the result. $\square$

Assumption checklist for applications. To apply Theorem 2.6, one must verify separately:

1. the full all-pairs graph inequality on a graph block containing the selected proximal graph points and the solution graph points;
2. pointwise Minty coverage and input-completeness for every input that the local orbit may visit;
3. the error bound for the same operator, residual, and solution set as those used by the proximal transition;
4. the residual-range condition in (L5);
5. at least one of the two strict scalar ratios; and
6. the positive localization margin (19).

Checking only RL cannot replace coverage, and checking the error bound for an auxiliary residual cannot replace item 3.

3 Native verification and a multivalued seam

This section separates three logically different questions which are easily conflated in applications: (i) whether the relevant Minty inputs are covered, (ii) whether the reflected graph satisfies the RL estimate, and (iii) whether the residual has the growth required by the error bound. The first two questions are genuinely all-branch questions, whereas the third is a one-point comparison with the zero set. The elementary calculus below is a way of importing branch formulas, derivative estimates, or error-bound criteria into the RL-EB theorem. It is not presented as a replacement for the established coderivative, slope, or higher-order subregularity theories which may be used to prove its hypotheses.

For a fixed $\lambda > 0$, write

$$M_\lambda^+(u, u^*) := u + \lambda u^*, \quad M_\lambda^-(u, u^*) := u - \lambda u^*.$$

Within this section, $\text{RL}(\gamma, L)$ abbreviates the condition $\text{RL}(\lambda, \omega)$ of Definition 2.1 with $\omega(t) = Lt^\gamma$; the proximal parameter λ is fixed.

Proposition 3.1 (Finite-chart RL verification). *Let $\mathcal{G} \subset \text{gph } F$ and let $U \subset \mathbb{R}^n$. Suppose that, for finitely many sets $U_i \subset U$ and mappings $T_i : U_i \rightarrow \mathbb{R}^n$,*

$$\text{gph } F \cap (M_\lambda^+)^{-1}(U) = \mathcal{G} \cap (M_\lambda^+)^{-1}(U) = \bigcup_{i=1}^N \left\{ \left(T_i z, \frac{z - T_i z}{\lambda} \right) : z \in U_i \right\}. \quad (3.1.1)$$

Set $R_i := 2T_i - I$. Assume the overlap consistency condition

$$T_i z = T_j z \quad \text{whenever } z \in U_i \cap U_j. \quad (3.1.2)$$

Thus the atlas is assumed to contain every graph point whose Minty input lies in U , not merely a selected family of branches. Then the Minty map M_λ^+ is one-to-one on the displayed graph block. If, in addition, $U = \bigcup_i U_i$, then the ordinary local resolvent of F exists and is single-valued on all of U , with $J_{\lambda F}|_{U_i} = T_i$.

For $0 < \gamma \leq 1$, the same graph block is $\text{RL}(\gamma, L)$ if and only if

$$\|R_i z - R_j w\| \leq L \|z - w\|^\gamma \quad (z \in U_i, w \in U_j, 1 \leq i, j \leq N). \quad (3.1.3)$$

In particular, branchwise estimates alone do not establish RL .

There is a useful gluing sufficient condition for (3.1.3). Suppose that every R_i is (γ, L_0) -Hölder on U_i . Suppose also that any $z \in U_i$ and $w \in U_j$ can be joined by points $z = z_0, \dots, z_m = w$, with $m \leq M$, such that each consecutive pair lies in one chart, consecutive charts agree at their common endpoint, and

$$\sum_{\ell=1}^m \|z_\ell - z_{\ell-1}\| \leq C \|z - w\|. \quad (3.1.4)$$

Then (3.1.3) holds with

$$L = L_0 M^{1-\gamma} C^\gamma. \quad (3.1.5)$$

Proof. For the point in the i -th chart, the two Minty coordinates are

$$M_\lambda^+ = z, \quad M_\lambda^- = 2T_i z - z = R_i z.$$

Thus overlap consistency is precisely the exclusion of a Minty collision, and the RL inequality is precisely (3.1.3). Under the gluing assumptions, telescope the reflected values along the chain and use Hölder continuity:

$$\|R_i z - R_j w\| \leq L_0 \sum_{\ell=1}^m \|z_\ell - z_{\ell-1}\|^\gamma \leq L_0 m^{1-\gamma} \left(\sum_{\ell=1}^m \|z_\ell - z_{\ell-1}\| \right)^\gamma.$$

Equations (3.1.4)–(3.1.5) follow. $\square$

Proposition 3.2 (Two-point order-ratio test). *Suppose the i -th branch is parametrized by graph points $g_i(\theta)$, and write*

$$\xi_i(\theta) := M_\lambda^+ g_i(\theta), \quad \zeta_i(\theta) := M_\lambda^- g_i(\theta).$$

Assume that the Minty-input block has diameter at most $D > 0$. For every ordered branch pair (i, j) , suppose there are $A_{ij}, c_{ij}, p_{ij}, q_{ij} > 0$ and a nonnegative comparator $\rho_{ij}(\theta, \eta)$ such that

$$\begin{aligned} \|\zeta_i(\theta) - \zeta_j(\eta)\| &\leq A_{ij} \rho_{ij}(\theta, \eta)^{p_{ij}}, \\ \|\xi_i(\theta) - \xi_j(\eta)\| &\geq c_{ij} \rho_{ij}(\theta, \eta)^{q_{ij}}. \end{aligned} \quad (3.2.1)$$

Set $r_{ij} := p_{ij}/q_{ij}$. For every

$$0 < \gamma \leq \min\{1, \min_{i,j} r_{ij}\},$$

the complete represented graph block satisfies $\text{RL}(\gamma, L)$ with

$$L = \max_{i,j} A_{ij} c_{ij}^{-r_{ij}} D^{r_{ij}-\gamma}. \quad (3.2.2)$$

The hypotheses also exclude Minty collisions.

Proof. Eliminating ρ_{ij} from (3.2.1) gives

$$\|\zeta_i(\theta) - \zeta_j(\eta)\| \leq A_{ij} c_{ij}^{-r_{ij}} \|\xi_i(\theta) - \xi_j(\eta)\|^{r_{ij}}.$$

Use the diameter bound to lower the exponent from r_{ij} to γ . If the Minty-input difference vanishes, the second inequality in (3.2.1) forces $\rho_{ij} = 0$, and the first then forces the reflected-output difference to vanish. $\square$

Remark 3.3. The order ratio p_{ij}/q_{ij} is especially useful at an active seam. It must be checked for adjacent *pairs* of branches; a one-branch Taylor expansion does not control a cross-branch collision. Conversely, two branch domains separated by a positive Minty-input gap require only a bounded cross-output estimate.

Proposition 3.4 (Independent residual-growth verification). *Let $S = F^{-1}(0) \cap D$ be closed, let $r > 0$, and let $\eta : [0, r] \rightarrow [0, +\infty)$ be nondecreasing, with $\eta(0) = 0$ and $\eta(s) > 0$ for $s > 0$. Suppose that*

$$\|y^*\| \geq \eta(d(y, S)) \quad \text{for every } (y, y^*) \in \text{gph } F \text{ under consideration with } d(y, S) \leq r. \quad (3.4.1)$$

Define the generalized upper inverse

$$\eta^\leftarrow(t) := \sup\{s \in [0, r] : \eta(s) \leq t\}.$$

Then

$$d(y, S) \leq \eta^\leftarrow(d(0, F(y))). \quad (3.4.2)$$

In particular, if $\eta(s) = ms^a$, then

$$d(y, S) \leq m^{-1/a} d(0, F(y))^{1/a}. \quad (3.4.3)$$

For a union of branches, hypothesis (3.4.1) must hold for every branch which enters the infimum defining $d(0, F(y))$.

Proof. Taking the infimum of (3.4.1) over $y^* \in F(y)$ gives $\eta(d(y, S)) \leq d(0, F(y))$. The definition of the generalized inverse then gives (3.4.2). The power case is immediate. $\square$

Corollary 3.5 (Two elementary EB assemblies). *The residual-growth hypothesis in Proposition 3.4 follows in either of the following situations.*

1. If $F(y) = \bigcup_{i \in I(y)} F_i(y)$ is a finite active union and

$$d(0, F_i(y)) \geq m_i d(y, S)^{a_i}$$

for every active branch, then on $d(y, S) \leq R$ one may take

$$a = \max_i a_i, \quad m = \min_i m_i R^{a_i - a}. \quad (3.5.1)$$

2. If, for every $v \in F(y)$, a known normal component satisfies

$$\|P_{N(y)} v\| \geq \eta(d(y, S)), \quad (3.5.2)$$

where $P_{N(y)}$ is an orthogonal projection, then one may use this η . In particular, for $y \notin S$ and $p \in P_S(y)$, it suffices that every value $v \in F(y)$ obey

$$\left| \left\langle v, \frac{y - p}{\|y - p\|} \right\rangle \right| \geq \eta(d(y, S)). \quad (3.5.3)$$

Proof. For a finite union the residual is the minimum of the active branch residuals. If $0 \leq t \leq R$ and $a_i \leq a$, then $t^{a_i} \geq R^{a_i-a} t^a$, which proves the first assertion. The second follows from $\|v\| \geq \|P_{N(y)}v\|$, followed by the infimum over all values of $F(y)$. $\square$

Remark 3.6 (What the verification does not allow one to omit). In a branch-defined problem, a complete certificate has the following order.

1. Enumerate the relevant solutions of $z = u + \lambda u^*$ on every active graph branch.
2. Check coverage of the intended input neighborhood.
3. On overlaps, exclude collisions: equal Minty input must give equal primal output.
4. Check both same-branch and cross-branch reflected estimates.
5. Separately establish residual growth against the entire zero set and for every value that can minimize the residual.
6. Only then compare the resulting (γ, L) and error-bound gauge in the scalar compatibility condition of the RL–EB theorem.

A coderivative theorem, a slope criterion, a polyhedral error bound, or a higher-order subregularity theorem may supply Step 5. Such a result does not by itself supply Steps 1–4; conversely, RL does not imply coverage or the required error bound.

We next give a closed-graph example in which all of these checks can be performed exactly. It is multivalued away from its nonisolated zero set, and the two Minty charts meet at a genuinely non-Lipschitz seam.

Theorem 3.7 (A two-branch square-root seam). *Define $F : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$ by*

$$F(\xi, y) = \begin{cases} \{(-2\sqrt{y}, 3y), (-2\sqrt{y}, -5y)\}, & y \geq 0, \\ \emptyset, & y < 0. \end{cases} \quad (3.7.1)$$

For the unit proximal parameter $\lambda = 1$, the following statements hold.

(a) *The graph is closed, the zero set is the nonisolated line*

$$S = F^{-1}(0) = \mathbb{R} \times \{0\}, \quad (3.7.2)$$

and the resolvent is globally defined and single-valued:

$$J_F(p, q) = \left(p + \sqrt{|q|}, \frac{|q|}{4} \right). \quad (3.7.3)$$

(b) *The reflected resolvent is*

$$Q := 2J_F - I, \quad Q(p, q) = \left(p + 2\sqrt{|q|}, \frac{|q|}{2} - q \right). \quad (3.7.4)$$

If $U \subset \mathbb{R}^2$ has diameter at most $D < +\infty$, then the corresponding complete graph block satisfies $\text{RL}(1/2, L_D)$, where the explicit valid constant

$$L_D := \left((2 + \sqrt{D})^2 + \frac{9}{4}D \right)^{1/2} \quad (3.7.5)$$

depends only on the Minty-input diameter.

(c) At every $(\bar{p}, 0) \in S$, the best local power exponent is exactly $1/2$. In particular, Q is not calm, hence not locally Lipschitz, at $(\bar{p}, 0)$. Consequently no finite local $\text{RL}(1, L)$ certificate is possible there.

(d) For $y \geq 0$, the residual is

$$r_F(\xi, y) := d(0, F(\xi, y)) = \sqrt{4y + 9y^2}, \quad (3.7.6)$$

and hence

$$d((\xi, y), S) = y \leq \frac{1}{4} r_F(\xi, y)^2. \quad (3.7.7)$$

The coefficient $1/4$ and the residual power 2 are asymptotically sharp. The exact gauge is

$$\psi_{\text{ex}}(t) = \frac{\sqrt{16 + 36t^2} - 4}{18}. \quad (3.7.8)$$

(e) For an arbitrary starting point $z^0 = (p_0, q_0)$, put $r_0 := |q_0|$. The PPA orbit $z^{k+1} = J_F z^k$ is

$$p_k = p_0 + 2\sqrt{r_0}(1 - 2^{-k}), \quad |q_k| = 4^{-k}r_0, \quad (3.7.9)$$

where $q_k = 4^{-k}r_0$ for every $k \geq 1$. Thus

$$d(z^{k+1}, S) = \frac{1}{4}d(z^k, S), \quad z^k \longrightarrow z^\infty := (p_0 + 2\sqrt{r_0}, 0). \quad (3.7.10)$$

The total length is finite. More precisely,

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\| \leq \begin{cases} 2\sqrt{r_0} + r_0, & q_0 \geq 0, \\ 2\sqrt{r_0} + \frac{3}{2}r_0, & q_0 < 0. \end{cases} \quad (3.7.11)$$

For $k \geq 1$, the exact pointwise tail is

$$\|z^k - z^\infty\| = \left(4r_0 4^{-k} + r_0^2 16^{-k}\right)^{1/2}, \quad (3.7.12)$$

so, when $r_0 > 0$, the pointwise asymptotic factor is $1/2$, while the distance-to-set factor is $1/4$. If $r_0 = 0$, the orbit is stationary.

Proof. The two graph branches have Minty inputs

$$\begin{aligned} (+) : \quad & (p, q) = (\xi - 2\sqrt{y}, 4y), \quad q \geq 0, \\ (-) : \quad & (p, q) = (\xi - 2\sqrt{y}, -4y), \quad q \leq 0. \end{aligned} \quad (3.7.13)$$

They cover the upper and lower half-planes, respectively. Their overlap is $q = 0$, where $y = 0$ and both graph values coincide. Thus there is no Minty collision, and solving (3.7.13) gives (3.7.3). Closedness follows directly from the two continuous branch parametrizations, and (3.7.2) is immediate.

Formula (3.7.4) follows from $Q = 2J_F - I$. Let $z = (p, q)$, $w = (p', q')$, and $\delta = \|z - w\|$. The inequalities

$$\left| \sqrt{|q|} - \sqrt{|q'|} \right| \leq \sqrt{|q - q'|} \leq \sqrt{\delta} \quad (3.7.14)$$

and

$$\left| \left(\frac{|q|}{2} - q \right) - \left(\frac{|q'|}{2} - q' \right) \right| \leq \frac{3}{2}|q - q'| \quad (3.7.15)$$

give

$$\|Qz - Qw\|^2 \leq (\delta + 2\sqrt{\delta})^2 + \frac{9}{4}\delta^2 \leq L_D^2\delta \quad (\delta \leq D). \quad (3.7.16)$$

This proves (b), including all cross-seam comparisons.

To see optimality across the two branches, take $z_t = (\bar{p}, t)$ and $w_t = (\bar{p}, -4t)$, where $t > 0$. Then

$$\|z_t - w_t\| = 5t, \quad \|Qz_t - Qw_t\| = \sqrt{4t + \frac{169}{4}t^2} \sim 2\sqrt{t}. \quad (3.7.17)$$

No local estimate with exponent $\alpha > 1/2$ can hold, and the quotient by $\|z_t - w_t\|$ diverges. This proves (c).

The smaller of the squared norms of the two values in (3.7.1) is $4y + 9y^2$, which proves (3.7.6). Solving $t^2 = 4y + 9y^2$ gives (3.7.8), whereas $r_F^2 \geq 4y$ gives (3.7.7). Moreover, $y/r_F^2 = (4 + 9y)^{-1} \rightarrow 1/4$. If an estimate $y \leq Cr_F^\theta$ held with $\theta > 2$, then $y/r_F^\theta \asymp y^{1-\theta/2} \rightarrow +\infty$; hence both assertions of sharpness follow.

Finally, (3.7.3) gives

$$q_{k+1} = \frac{|q_k|}{4}, \quad p_{k+1} = p_k + \sqrt{|q_k|}. \quad (3.7.18)$$

Summing the resulting geometric sequences proves (3.7.9)–(3.7.10). For $q_0 \geq 0$, the horizontal and vertical variations sum to $2\sqrt{r_0}$ and r_0 , respectively. If $q_0 < 0$, the first vertical variation is $5r_0/4$ and the remaining vertical variations sum to $r_0/4$. The triangle inequality yields (3.7.11), and direct subtraction of (3.7.9) from the limit gives (3.7.12). $\square$

Remark 3.8 (How the two certificates meet). On a sufficiently small, projection-stable Minty neighborhood, $L_D < 4$. If $d = d(z, S)$, $s = \|z - J_F z\|$, and $d_+ = d(J_F z, S)$, the RL step estimate and (3.7.7) give

$$s \leq \frac{1}{2}(d + L_D\sqrt{d}), \quad d_+ \leq \frac{1}{4}s^2,$$

and therefore

$$d_+ \leq \frac{(\sqrt{d} + L_D)^2}{16} d. \quad (3.8.1)$$

This is a direct RL–EB contraction certificate on a positive neighborhood. More precisely, take the RL scale and the input-block diameter both equal to R , and write

$$L_R = \left((2 + \sqrt{R})^2 + \frac{9}{4}R \right)^{1/2}.$$

Since $\psi(t) = t^2/4$ gives $h(d) = 4d$, the branch-(H) ratio of Theorem 2.6 is

$$q_H(R) = \frac{(\sqrt{R} + L_R)^2}{16} \longrightarrow \frac{1}{4} \quad (R \downarrow 0). \quad (3.8.2)$$

Thus the generic certificate is asymptotically sharp for the distance factor of this operator. For example, $R = 10^{-2}$ gives $q_H(R) < 0.305$, a strict positive-radius contraction certificate. The exact factor $1/4$ in (3.7.10) is stronger because this example is explicitly solvable at every scale; no class-wide optimality is claimed.

Remark 3.9 (Exact scope of the separation). Equation (3.7.17) rules out every local theory whose graph hypothesis entails a finite linear reflected-resolvent estimate. In particular, it rules out the $\gamma = 1$ RL/Luke–Tam slice and the balanced quadratic semimonotonicity slice

equivalent to it. The example is also not locally hypomonotone with any finite modulus at any seam point. Fix $\bar{p} \in \mathbb{R}$ and compare

$$u_t = (\bar{p} + t, t), \quad u_t^* = (-2\sqrt{t}, 3t) \quad \text{with} \quad (\bar{p}, 0; 0, 0).$$

The comparison gives

$$\frac{\langle u_t - (\bar{p}, 0), u_t^* \rangle}{\|u_t - (\bar{p}, 0)\|^2} = -\frac{1}{\sqrt{t}} + \frac{3}{2} \longrightarrow -\infty. \quad (3.9.1)$$

The same concrete map also fails two nearby solution-anchored certificates on every full neighborhood of $(\bar{p}, 0)$. First, with $z_t = (\bar{p}, t)$ and $\bar{z} = (\bar{p}, 0)$,

$$\frac{\|J_F z_t - J_F \bar{z}\|}{\|z_t - \bar{z}\|} = \frac{\sqrt{t + t^2/16}}{t} \longrightarrow +\infty. \quad (3.9.2)$$

Hence the same-map, fixed-anchor almost-nonexpansive—and therefore almost-averaged—test with finite violation from [12] cannot hold there. Second, choose

$$u_t = (\bar{p} + t^{1/4}, t), \quad u_t^* = (-2\sqrt{t}, 3t) \in F(u_t).$$

Then

$$\frac{\langle u_t - (\bar{p}, 0), u_t^* \rangle}{\|u_t^*\|^2} = \frac{-2t^{3/4} + 3t^2}{4t + 9t^2} \longrightarrow -\infty. \quad (3.9.3)$$

Consequently no fixed finite residual-quadratic anchored inequality $\langle u - \bar{u}, u^* \rangle \geq \langle u^*, V u^* \rangle$ can hold on that graph neighborhood, including the corresponding weak/oblique-Minty certificate of [2] with fixed finite V . These exclusions concern the same operator, anchor, residual, and full neighborhood. They do not exclude a different domain, a partial or degenerate test, a changed preconditioner, or a model-specific argument.

Remark 3.10 (Scope of the example). The multifunction F is genuinely two-valued for $y > 0$, but its resolvent is single-valued because the two complete Minty images occupy opposite half-planes. Thus the example concerns a deterministic multivalued graph seam, not random branch selection or a set-valued PPA step. The pure 1/2-power RL estimate is local: no fixed pure-power constant controls the unbounded p -direction globally. Finally, the explicit recursion proves convergence directly. The role of the example is to demonstrate that RL-EB reaches a natural non-Lipschitz, nonisolated-solution geometry outside its linear reflected-resolvent slice; it does not show that no other problem-specific argument can prove the same orbit converges.

4 Relation to existing generalized monotonicity and regularity theories

4.1 The linear RL slice is established quadratic geometry

The exponent-one case should first be calibrated against existing notions. For two graph differences $a = u - v$ and $b = u^* - v^*$, set $c = \lambda b$. Expanding the squares gives the exact equivalence

$$\|a - c\| \leq L\|a + c\| \iff \langle a, c \rangle \geq \frac{1 - L^2}{2(1 + L^2)} (\|a\|^2 + \|c\|^2). \quad (37)$$

In the unscaled graph coordinates, (37) is the balanced slice of (μ, ρ) -semimonotonicity introduced in [2], with

$$\mu = \frac{1 - L^2}{2\lambda(1 + L^2)}, \quad \rho = \lambda^2 \mu. \quad (38)$$

When $L > 1$, the scaled inequality is also precisely the ϱ -semimonotonicity of [3], with $\varrho = (L^2 - 1)/(L^2 + 1)$. The endpoint $L = 1$ is ordinary monotonicity.

The same identity recovers the Luke–Tam normalization. Their scaled submonotonicity inequality has the form

$$\langle a, c \rangle \geq -\tau \|a + c\|^2,$$

and hence

$$L^2 = 1 + 4\tau, \quad \frac{1 + L^2}{2} = 1 + 2\tau. \quad (39)$$

Thus the case $\gamma = 1$ of our condition is not proposed as a new operator class. Its purpose is to connect the nonlinear theory continuously to monotonicity, semimonotonicity, and submonotonicity-based PPA analyses [11].

Hypomonotonicity is another sufficient source of exponent-one RL parameters. Indeed, if $h \geq 0$ and

$$\langle a, b \rangle \geq -h \|a\|^2 \quad \text{and} \quad \lambda h < 1,$$

then a direct quadratic calculation yields

$$\|a - \lambda b\| \leq \frac{1 + \lambda h}{1 - \lambda h} \|a + \lambda b\|. \quad (40)$$

For subdifferential mappings, prox-regularity and attentive localization give standard local hypomonotonicity and Lipschitz proximal behavior under the corresponding stepsize restriction [14]. Such results are therefore verification tools for the linear slice; without additional high-order analysis, they do not certify an optimal exponent below one.

4.2 Error bounds are imported as an independent module

Metric and Hölder subregularity already have extensive variational-analytic theories. Li and Mordukhovich define metric q -subregularity for $q > 0$, while their main Hölder results and PPA applications focus on $q \in (0, 1]$ for maximal monotone mappings [10]. Mordukhovich and Ouyang develop metric and strong metric subregularity of arbitrary positive order, including $q > 1$ [13]. Kruger’s slope framework supplies primal and dual criteria for linear and nonlinear subregularity [6, 7], whereas admissible-function subregularity and coderivative conditions are available from [16]. More recently, Li, Mordukhovich, and Zhu derived in their Theorems 4.3–4.4 an abstract generalized-subregularity framework with finite-length and nonlinear-rate conclusions for objective sequences satisfying descent, surrogate-step, and relative-error assumptions [9]. Their input is therefore a different algorithmic sequence package, rather than an all-pairs Minty graph condition for a proximal transition.

Accordingly, neither a general gauge nor a higher-order error bound is claimed as new here. These theories provide independent certificates for our hypothesis

$$\text{dist}(y, S) \leq \psi(\text{dist}(0, F(y))).$$

For example, a power error bound

$$\text{dist}(y, S) \leq K \text{dist}(0, F(y))^q \quad (41)$$

corresponds to $\psi(t) = Kt^q$. In the genuine nonlinear regime $0 < \gamma < 1$ and $L > 0$, coupling (41) with an RL modulus Ld^γ exposes the order $d^{\gamma q}$: the scalar certificate is automatically order-compatible when $\gamma q > 1$, becomes a constant-critical condition when $\gamma q = 1$, and cannot yield a fixed strict contraction when $\gamma q < 1$. The last statement concerns this certificate, not the behavior of every orbit. It is not asserted for $\gamma = 1$ with $L < 1$, where the RL energy can itself be contractive.

The general gauge is therefore used as an abstract interface rather than as the main novelty claim. The currently verified flagship application and the convergence illustration in Example 4.6 of [9] use power gauges. That paper also gives a genuinely non-power regularity construction in Example 3.3, but it is not a proximal-orbit application of the present geometry. Unless a genuinely non-power gauge is exhibited whose local orbit behavior is not already captured by a power envelope, the substantive applications should be positioned through the sharp power-law compatibility and the multibranch junction analysis.

Ordinary metric subregularity in (41) measures distance to the full inverse set and therefore allows nonisolated solutions. It should not be replaced silently by strong metric subregularity, whose local form singles out the reference solution. Generalized-differential criteria for constraint and implicit multifunctions can also verify subregularity, calmness, local inverse properties, or coverage [4, 5]. Coderivative characterizations of maximal monotonicity provide a complementary route to classical graph geometry [1]. A pointwise or directional test, however, must still be upgraded to a uniform statement on the complete graph block before it can establish the all-pairs RL condition.

4.3 Rate calibration

At the monotone endpoint $L = 1$, the coupled energy and linear error-bound calculation gives

$$\text{dist}(x^+, S) \leq \frac{\rho}{\sqrt{\rho^2 + \lambda^2}} \text{dist}(x, S), \quad (42)$$

which is precisely the distance factor in Leventhal’s Theorem 3.1 [8]. Thus (42) is a consistency check, not a new monotone-PPA rate.

For the Luke–Tam normalization $L^2 = 1 + 4\tau$, retaining the output error bound at x^+ and coupling it directly with the energy inequality yields

$$\kappa_{\text{coupled}}^2 = (1 + 2\tau) \frac{\rho^2}{\rho^2 + \lambda^2}, \quad \kappa_{\text{coupled}} < 1 \iff 2\tau\rho^2 < \lambda^2. \quad (43)$$

Luke and Tam instead convert subregularity into an input-step estimate and obtain in their proof

$$\kappa_{\text{LT}}^2 = 1 + 2\tau - \left(\frac{\lambda}{\lambda + \rho} \right)^2,$$

together with the sufficient compatibility $\tau(1 + \rho/\lambda)^2 < 1/2$ [11]. Formula (43) is a sharper consequence of the shared energy inequality and the output error bound. We do not infer dominance over their complete local theorem, whose self-mapping, maximal-local-submonotonicity, coverage, and selection hypotheses form a different theorem package.

4.4 The genuine nonlinear regime and its boundary

Almost-averaged fixed-point theory combines a finite quadratic violation with metric subregularity of an iteration residual and gives broad quantitative local convergence results [12]. Its local convergence theorem uses pointwise almost averagedness at solution anchors, not an all-pairs Lipschitz estimate over the entire neighborhood. Oblique weak-Minty PPA theory instead compares graph points with one or more selected zeros and accommodates positive-semidefinite preconditioning [2]. The hypotheses encode different information: the former is a fixed-point-map framework, the latter uses a solution-anchored graph inequality, and RL uses every pair of graph points on a prescribed block. We claim no general inclusion between the complete frameworks.

The distinct regime considered here occurs when the complete reflected resolvent is Hölder across a multivalued branch junction but is not locally Lipschitz there. In that case no finite exponent-one RL constant, and hence no Luke–Tam quadratic certificate with finite violation, exists on the same full graph block. This statement is deliberately local and certificate-specific. In particular, failure of the all-pairs Lipschitz bound does not by itself exclude the pointwise solution-anchored condition of [12], nor does it preclude a weak-Minty argument, a different domain or residual, or the use of additional problem structure. For the square-root seam, Remark 3.9 goes further: separate explicit sequences rule out the same-map fixed-anchor almost-averaged test and every fixed finite residual-quadratic anchored weak-Minty inequality on the same complete neighborhood.

The bounded contribution of the present framework is therefore the quantitative pairing of this all-pairs, non-Lipschitz graph geometry with an independently verified output error bound for the same proximal transition, plus explicit Minty coverage and a finite-length localization argument. A valid strict example must prove both a Hölder upper estimate and a matching sequence that excludes every larger exponent; merely rewriting a Lipschitz estimate with a smaller exponent would not establish a genuinely nonlinear case.

5 Conclusion

The useful content of RLEB is not a new name for Hölder continuity and not a replacement for established error-bound theory. It is a graph-level way to retain every within-branch and cross-branch comparison needed by a proximal transition, and then to couple that nonlinear reflected geometry with an independently certified residual-to-solution estimate. The resulting theorem keeps three logically different requirements visible: Minty injectivity, input coverage, and residual recovery. Under either of two explicit scalar compatibility branches, these ingredients yield Lyapunov decay, a finite localization budget, finite length, and convergence of the full proximal orbit to one point of a possibly nonisolated solution set.

The square-root seam shows that this distinction is substantive. Its complete reflected resolvent has optimal exponent $1/2$ and no finite local Lipschitz constant, while it has the matching quadratic error bound needed by the power-law theorem. Hence the complete multivalued junction is certified even though the corresponding exponent-one all-pairs quadratic certificate fails on the same graph block. At the same time, the example does not imply that all solution-anchored, partial, or problem-specific methods fail; the comparison is deliberately certificate-specific.

The remaining research problem is therefore precise. One must develop native, quantitative ways to verify complete cross-branch moduli, coverage, and output error bounds for broader structured multifunctions without losing their dependence. The finite-chart and order-ratio rules given here are elementary assembly tools for that task, not yet a general derivative/coderivative verification theory. Extending the theory beyond finite-dimensional Euclidean space, and identifying non-power gauges whose orbit behavior is not captured by a local power envelope, are separate questions left open.

References

- [1] Nguyen Huy Chieu, Gue Myung Lee, Boris S. Mordukhovich, and Tran T. A. Nghia. Coderivative characterizations of maximal monotonicity for set-valued mappings. *Journal of Convex Analysis*, 23(2):461–480, 2016.

- [2] Brecht Evens, Pieter Pas, Puya Latafat, and Panagiotis Patrinos. Convergence of the preconditioned proximal point method and Douglas–Rachford splitting in the absence of monotonicity. *Mathematical Programming*, 214:247–301, 2025.
- [3] Rolando Gárciga Otero and Alfredo Iusem. Regularity results for semimonotone operators. *Computational & Applied Mathematics*, 30(1):5–17, 2011.
- [4] Helmut Gfrerer. First order and second order characterizations of metric subregularity and calmness of constraint set mappings. *SIAM Journal on Optimization*, 21(4):1439–1474, 2011.
- [5] Helmut Gfrerer and Jiří V. Outrata. On lipschitzian properties of implicit multifunctions. *SIAM Journal on Optimization*, 26(4):2160–2189, 2016.
- [6] Alexander Y. Kruger. Error bounds and metric subregularity. *Optimization*, 64(1):49–79, 2015.
- [7] Alexander Y. Kruger. Nonlinear metric subregularity. *Journal of Optimization Theory and Applications*, 171(3):820–855, 2016.
- [8] D. Leventhal. Metric subregularity and the proximal point method. *Journal of Mathematical Analysis and Applications*, 360(2):681–688, 2009.
- [9] Guoyin Li, Boris Mordukhovich, and Jiangxing Zhu. Generalized metric subregularity with applications to high-order regularized newton methods. *Mathematics of Operations Research*, 2026. Published online 7 April 2026; official cite-as record was volume 0, number 0 at the 2026-09-17 audit.
- [10] Guoyin Li and Boris S. Mordukhovich. Hölder metric subregularity with applications to proximal point method. *SIAM Journal on Optimization*, 22(4):1655–1684, 2012.
- [11] D. Russell Luke and Matthew K. Tam. Generalized monotonicity and the proximal point algorithm. *Mathematics of Operations Research*, 2025. Published online 21 November 2025; issue, volume, and pages not assigned on the official page at the 2026-09-17 audit.
- [12] D. Russell Luke, Nguyen H. Thao, and Matthew K. Tam. Quantitative convergence analysis of iterated expansive, set-valued mappings. *Mathematics of Operations Research*, 43(4):1143–1176, 2018.
- [13] Boris S. Mordukhovich and Wei Ouyang. Higher-order metric subregularity and its applications. *Journal of Global Optimization*, 63(4):777–795, 2015.
- [14] R. A. Poliquin and R. Tyrrell Rockafellar. Prox-regular functions in variational analysis. *Transactions of the American Mathematical Society*, 348(5):1805–1838, 1996.
- [15] R. Tyrrell Rockafellar. Monotone operators and the proximal point algorithm. *SIAM Journal on Control and Optimization*, 14(5):877–898, 1976.
- [16] Xi Yin Zheng and Jiangxing Zhu. Generalized metric subregularity and regularity with respect to an admissible function. *SIAM Journal on Optimization*, 26(1):535–563, 2016.